

MOTIVATION

Challenge: Various applications across multiple screens, inconsistent visual designs and constant alerts overload operators and strain efficiency and decision-making under pressure

Goal: Design human-centred visualisations that can better support situational awareness and decision-making in control rooms

METHODOLOGY

1.Understanding Current Control Room Practice: Observations

1 Day Training Session

2 Operators Differing expertise: one operator and one specialist

3 ~1 hour scenarios Simulate contrasting conditions involving different applications and procedures

1 Day Live Control Room Observation

- observation notes and sketches
- audio recording of short interviews with operators

Data collected:

- Eye tracking gaze and video using Tobii Pro 3 glasses
- Videos of workstation and room
- Workload and situational awareness questionnaires
- 270 timestamped observer notes from 6 researchers with diverse expertise
- Control room layout and scenario information
- A list of applications and interfaces (name and purpose)



Figure 1. Photos from the observation of training session. Left: the setup of our cameras on tripods for capturing operator's workspace. Right (top and bottom): Operators introduced applications to researchers. Right (middle): The Tobii Pro 3 eye tracking glasses we used for collecting gaze data.

2. Understanding Current Control Room Practice: Analysis

Eye Tracking Visual Analysis using Gazealytics [1] Eye tracking data from 2 participant in 1 scenario, capturing ~6,000 fixations and saccades per participant

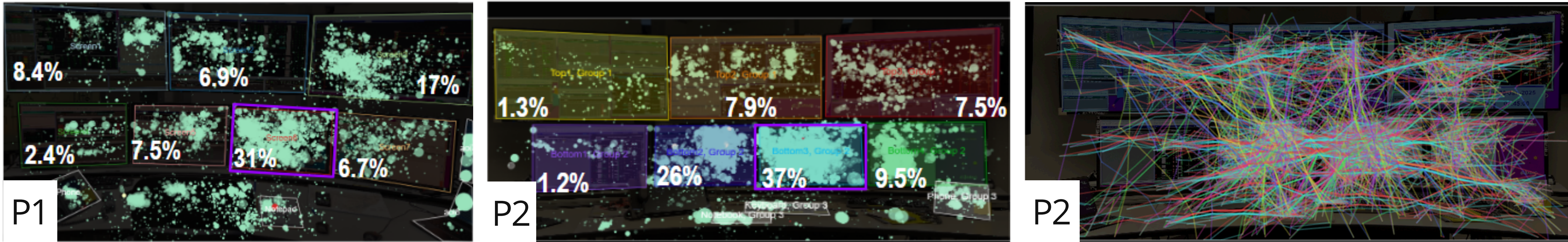


Figure 2. Screenshots of the visual analysis of eye tracking data using Gazealytics. Left: The fixations distribution of P1 with most fixations on Screen 6 and second on Screen 3. Middle: The fixations distribution of P2 with most fixations on Screen 6 and second on Screen 5. Right: The saccades distribution of P2 with eye movements in all directions.

Initial findings:

- Eye movements occur in all directions (predominantly left to right) across the full workspace
- Majority of fixations on Screen 6 (the third bottom screens from the left) for both P1 (31%) and P2 (37%)
- Distinct patterns of other screen use between the two participants

Some screens with high percentages (particularly Screens 6, 5, 7) display multiple applications over time

Task and Application Analysis

17 applications and many more interfaces for 1 participant in 1 scenario

Rapid decision-making involves opening/closing and constant switching between applications/interfaces

Currently analysing temporal AOIs (Area of Interest)...

An example of identified key tasks ➡

Alarm investigation and verification task

2.5 Minutes **4 Applications**

10 Interfaces **14 Switches**

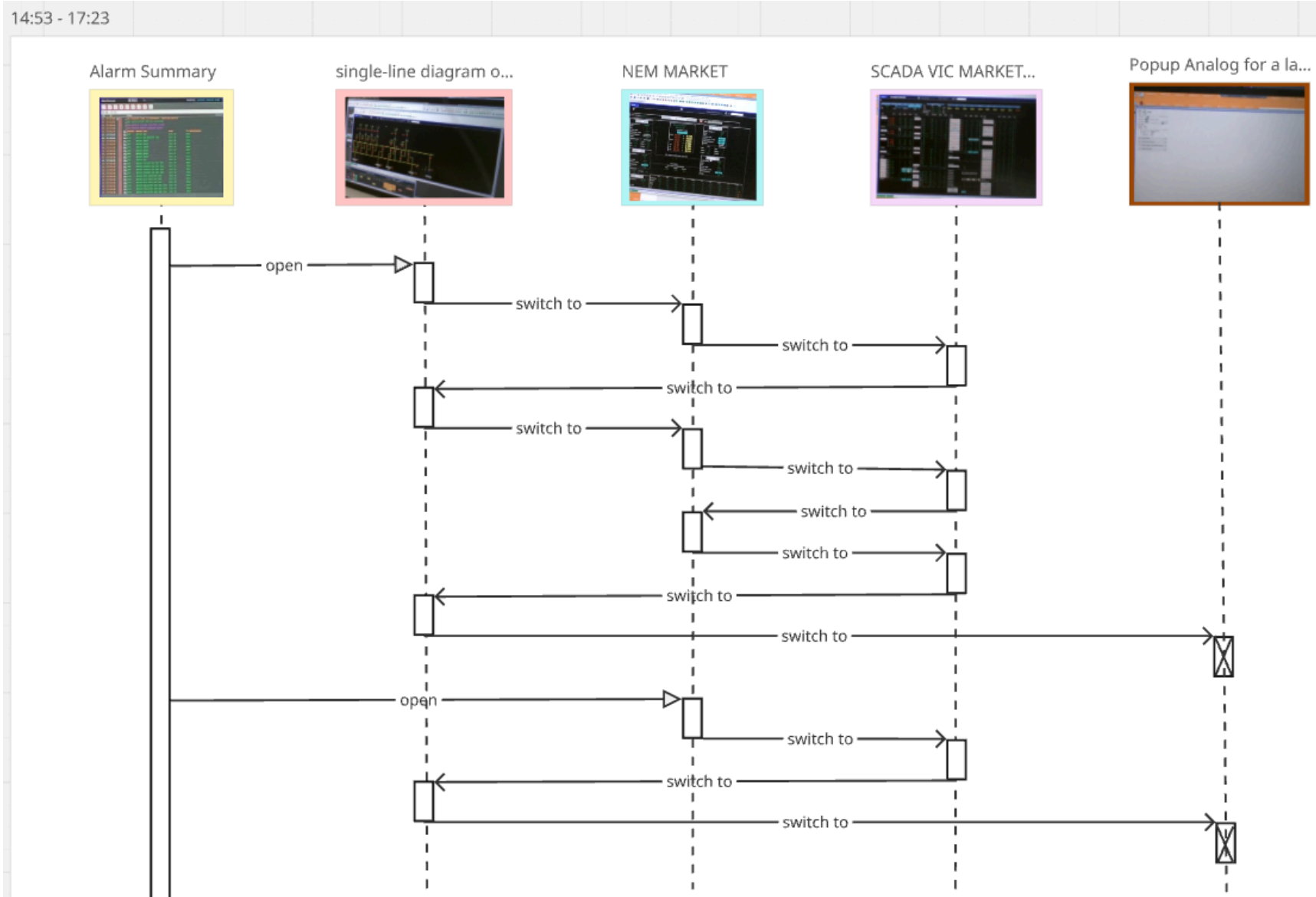


Figure 3. A timeline diagram of a 2.5 minute alarm investigation and verification task showing frequent switching across applications and interfaces.

3. Understanding Future Control Room Needs: Ideation Workshop

 **16 Participants**

- Energy control room: operation, training and software support
- Data visualisation
- Decision support
- AI & Optimisation

4 Activities

3 Initial insights

- **Imagine automated, streamlined, and reliable systems with unified interface, smarter search, and AI assistance:**
- Trial intelligent systems for well-defined tasks, e.g. handovers, alarm filtering, and procedure recommendations
- Keep human in charge: AI should support, never replace human judgement
- Redesign interfaces: Streamline information, highlight and prioritise what matters most

Sketchy Icebreaker

Wishful Thinking

Barrier Removal

Storyboarding

Figure 4. Artefacts produced from the Ideation Workshop. 1) Introduction and icebreaker: Participants were asked to introduce their names and positions, and answer a ice-breaking question 2) Wishful thinking: Participants were encouraged to explore a wide idea space of what they would like to Know, See and Do. 3) Barrier removal: Identify the barriers for achieving the ideas from activity 2 4) Storyboarding: Participants worked in small groups to mocked up future control rooms.

CURRENT & FUTURE WORK

Continued analysis of temporal AOIs and interface switching activities (in progress)

4. Workshop with wider control room sector (to be completed)

5. Recommendations and Guidelines (to be completed)